INCH-POUND

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MILITARY SPECIFICATION
INTERROGATOR SET AN/TPX-46A(V)7

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for the construction and testing of an Identification Friend or Foe Interrogator Set designated AN/TPX-46A(V)7.

2. APPLICABLE DOCUMENTS.

2.1 Issues of Documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) issues different from those specified in the cited specifications are set forth in the solicitation. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-11268	- Parts, Materials and Processes used in Electronic Equipment.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes for Ground Electronic Equipment.
MIL-F-15024/6	- Plate, Identification, Information and Marking.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TM, Fort Monmouth, New Jersey 07703-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5895

AMSC - N/A

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STANDARDS

MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-2000	- Standard Requirements for Soldering Electrical and Electronic Assemblies.
MIL-STD-454	-Standard General Requirements for Electronic Equipment.
MIL-STD-461	-Electromagnetic Emission and Susceptibility Requirement for the Control of Electromagnetic Interference.
MIL-STD-462	-Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-781	-Reliability Tests Exponential Distribution.
MIL-STD-810	-Environmental Test Methods.
MIL-STD-1472	-Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1562	-List of Standard Microcircuits.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094 or as directed by the contracting officer.)

DOD AIMS SYSTEM PROGRAM OFFICE

DOD AIMS 64-900	- Mode 4 Input/Output Data KIR/KIT-1A/TSEC.

(A copy of DOD AIMS 64-900 required by contractors in connection with specific procurement functions should be obtained from the AIMS Program Office, Warner Robins Air Logistics Center, Warner Robins Air Force Base, GA 31098 or as directed by the contracting officer.)

DRAWINGS.

C5080975	-Interrogator Set AN/TPX-46A(V)7.
5071814	-Test Requirements for Interrogator Set.
SM-A-787975	-Test Requirements for Control, Interrogator Set.

(Copies of drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this document takes precedence. Nothing in this document, however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS.

3.1 First Article. When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

3.2 Construction. The AN/TPX-46A(V)7 shall be constructed in accordance with the requirements of this specification, C5080975, and the following list of major components:

Component	Drawing
Signal Processor, CP-1187B/TPX-46A(V)7	A3154738
Control Interrogator Set C-9961/TPX-46A(V)7	SM-E-787795
Cable assy, special purpose control panel	SM-D-787797
Cable assy, dc power	SM-D-787808
Cable assy, ac power	SM-D-787809
Installation Kit, program connector MK-2664/TPX-46A(V)7	A3154791

3.2.1 Government furnished equipment (GFE). If required by contract, the following government furnished equipment shall be provided:

a. Computer KIR-1A/TSEC with Power Supply ZACA/TSEC and its characteristics in DOD AIMS 64-900 when supplied with primary power input of 115 Volts AC, 400 HZ.

b. Code Key KIK-18/TSEC.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply to microcircuits selected from MIL-STD-1562, and Soldered Electrical and Electronic Assemblies shall be per MIL-STD 2000 (see 4.4).

3.3.1 Recycled virgin and reclaimed materials. It is requested that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin, shall be used without jeopardizing the intended use of the manufactured items.

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3.3.2 Substitutability. This requirement establishes criteria for the selection and application of substitute parts in accordance with MIL-STD-454, Requirement 72. Requirement 72 is applicable for production use by the contractor, but shall not be used for initial qualification/reliability demonstration, First Article, or initial production lot testing.

3.4 Finish. All surfaces of each unit part of AN/TPX-46A(V)7 requiring a protective coating shall be finished in accordance with MIL-F-14072 and the equipment drawings specified herein (see 4.4).

3.5 Nameplate and marking.

3.5.1 General. Marking of AN/TPX-46A(V)7 shall be in accordance with MIL-M-13231.

3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.5.3 Nameplates. Each nomenclatured item shall have a nameplate conforming to MIL-P-15024/6, Type H or B (see 4.4.).

3.5.4 Serial numbers. The following components of AN/TPX-46A(V)7 shall be so serial numbered (see 4.4):

Signal Processor, CP-1187B(V)/TPX-46A(V)7.

Control, Interrogator Set, C-9661/TPX-46A(V)7.

3.6 Safety engineering.

3.6.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1 shall be made applicable for the tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. The requirements of MIL-STD-1472, paragraph entitled "Labeling, Hazards and Safety" apply. Verification that compliance with the requirement has been achieved shall be through a visual examination and inspection (see 4.13).

3.6.2 Rounding exposed edges. Exposed corners and edges shall be rounded sufficiently to preclude puncture or laceration hazards.

3.6.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, and lenses.)

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3.7 Interchangeability.

3.7.1 Mechanical interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be functionally interchangeable without modification and shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.7.2 Electrical interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be functionally interchangeable without modification and shall conform to Requirement 7 of MIL-STD-454. Electrical adjustments may be made at all maintenance levels, as referenced in the manuals (see 4.8.).

3.8 Dimensions and weights. The maximum equipment dimensions and weights shall be listed in table I.

TABLE I. Dimensions and weights

Equipment	Height (in)	Depth (in)	Width (in)	Unit weight (lb)
Signal Processor, CP-1187B/TPX-46A(V)7	21.5 (54.6 cm)	22.12 (56.2 cm) (Excluding handles)	9.75 (24.3 cm)	85.0 (38.63 kg)
Control Interrogator Set, C-9661/TPX-46A(V)7	8.12 (20.6 cm)	16.0 (40.6 cm)	10.0 (25.4 cm)	10.0 (4.55 kg)

3.9 Burn-in/Pre-conditioning. The major component parts of the AN/TPX-46A(V)7 shall meet the performance requirements specified herein after subjection to vibration and electrical preconditioning tests of 4.6.

3.10 Performance requirements. The AN/TPX-46A(V)7 shall meet the performance requirements specified in 3.2 after subjection to the test of 4.9.

3.11 Service conditions. The AN/TPX-46A(V)7 shall meet the following service conditions (see 4.10).

3.11.1 Altitude. The AN/TPX-46A(V)7 shall meet specified performance, without degradation, at altitudes up to and including 10,000 feet (3048m) above sea level and shall withstand air transportation at altitudes up to 40,000 feet (12,192m) above sea level (see 4.10.1).

3.11.2 High temperature. The AN/TPX-46A(V)7 shall be operable without deterioration damage or degradation in specified performance at ambient air temperature as high as +125°F (51.67°C) plus the effect of solar radiation at a rate of 360 BTU/ft^2 /hr. The AN/TPX-46A(V)7 shall be permitted a 30 minute cool-down period before full operation is required. At the end of the 30

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minute period, the AN/TPX-46A(V)7 shall provide full specified performance. The AN/TPX-46A(V)7 shall be capable of withstanding, without degradation of specified performance, long periods of storage at temperatures as high as +160°F (71.1°C) (see 4.10.2).

3.11.3 Low temperature. The AN/TPX-46A(V)7 shall be operable without deterioration damage or degradation in specified performance at air temperatures as low as -50°F (-45.6°C). The AN/TPX-46A(V)7 shall be permitted a 30 minute warm-up period; it may be turned on as desired to benefit overall operational performance. At the end of the 30 minute period, the AN/TPX-46A(V)7 shall provide full specified performance. The AN/TPX-46A(V)7 shall be capable of withstanding, without degradation of specified performance, long periods of storage at room temperature as low as -70°F (-56.7°C) (see 4.10.3).

3.11.4 Humidity. The AN/TPX-46A(V)7 shall be operable without degradation in specified performance and shall sustain no physical damage, during and after prolonged exposure to extreme humidity levels as encountered in tropical areas (see 4.10.4).

3.11.5 Salt fog. After the salt fog test of 4.10.5, the AN/TPX-46A(V)7 shall operate without degradation in specified performance and show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface exfoliation. There shall also be no clogging or binding of moving mechanical parts.

3.11.6 Fungus. After the fungus test of 4.10.6, the AN/TPX-46A(V)7 shall be visually examined. the AN/TPX-46A(V)7 shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than six unrelated minuted colonies with mycelial development in areas only in other than critical circuit portions, such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the AN/TPX-46A(V)7 shall operate without degradation in the manufacture, fabrication, construction and treatment of the AN/TPX-46A(V)7 shall comply with Requirement 4 of MIL-STD-454.

3.11.7 Rain. The AN/TPX-46A(V)7 shall be operable without degradation in the specified performance after exposure to precipitation at variable rates to a total of 9.5 inches over a twelve hour period with a horizontal wind velocity of 35 knots per hour (see 4.10.7).

3.11.8 Dust. The AN/TPX-46A(V)7 shall be operable without degradation in specified performance after being subjected to the dust test of 4.10.8.

3.11.9 Vibration, sinusoidal. The AN/TPX-46A(V)7 shall operate within specified performance after being subjected to vibration test specified in 4.10.9.

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3.11.10 Shock. The AN/TPX-46A(V)7 shall operate within specified performance after being subjected to the method and condition of the shock test specified in 4.10.10.

3.11.11 Bench-handling. The AN/TPX-46A(V)7 shall withstand the shock as encountered during servicing. The equipment shall withstand this test removed from its normal enclosure as it would appear on the repair level (see 4.10.11).

3.12 Electromagnetic interference (EMI). The equipment shall meet the following emission and susceptibility requirements of Notice 4 to MIL-STD-461A when tested in accordance with the requirements of 4.11:

CE01	CS01	(b) RE02		RS02
CE02	CS02	(b) RE02.1	(e)	RS03
CE03	CS03		(e)	RS03.1
(a) CE04	(d) CS04			
(a) CE05	CS06			
(c) CE06				

a. Conducted switching transients shall comply with the limits of Figure 5 of Notice 4 to MIL-STD-461A. Paragraphs 6.2.1 in Notice 4 to MIL-STD-461A is to be consulted with regard to explanation/definition only.

b. Radiated switching transients shall comply with the requirements of paragraph 6.13.4 of Notice 4 to MIL-STD-461A.

c. In the 'key-up' (non-transmitting) condition, the requirements of paragraph 6.4 of Notice 4 to MIL-STD-461A apply except that, at the local oscillator frequency, the emissions shall not be greater than -67 dBmW. In 'key-down' (transmitting) condition, the requirements of paragraph 6.4.3 of Notice 4 to MIL-STD-461A apply with the exception that at 2fo, 3fo and 4fo (where fo is the transmitted fundamental frequency) the limits are relaxed by 10 dB.

d. The spurious response rejection level shall be greater than or equal to 80 dB.

e. Requirement RS03/RS03.1 requirement applies over the frequency range of 10 kHz to 12.4 GHz (inclusive) using the following field intensity levels:

10 kHz to 1.999 MHz - 1 volts-per-meter
2 MHz to 1.999 GHz - 50 volts-per-meter
2 GHz to 12.40 GHz - 5 volts-per-meter

3.13 Reliability. The AN/TPX-46A(V)7 shall have a specified (θ_o) mean-time-between-failure (MTBF) of 450 hours the lower test MTBF (θ_1) shall be 225 hours. The reliability requirements shall be demonstrated under the conditions specified in 4.12.

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3.14 Workmanship. The AN/TPX-46A(V)7 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454, the applicable portions of the following paragraphs in MIL-P-11268, and Soldered Electrical and Electronic Assemblies per MIL-STD-2000 (see 4.13):

General requirements for plastic
Wiring and cabling, including:
Slack
Protection
Clearance
Splicing
Connections, general
Grounding, general
Shielding on wire and cable
Brazing
General requirements for securing of parts
Cleaning prior to welding
Process of welding
Controls
Self-tapping screws
Cleaning after welding

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)

b. Inspections covered by subsidiary documents (see 4.4)

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First article: Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish five first article units of the AN/TPX-46A(V)7 for group A, B, C and D testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table II and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

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TABLE II. First article inspection.

Inspections	Reqt para	Test para	Order of Tests			
			Unit 1	Unit 2	Unit 3	Units 4-5
1. Inspection covered by subsidiary documents 1/	3.3 3.4 3.5	4.4	Inspection to be performed on all units.			
2. Burn-in/Precondition 1/	3.9	4.5	Tests to be performed on all units.			
3. Group A 1/	See Table III.		Tests to be performed on all units.			
4. Group B 1/	See Table IV.		Tests to be performed on all units.			
5. Group C.						
Altitude	3.11.1	4.10.1	1			
Low temperature	3.11.3	4.10.3	2			
High temperature	3.11.2	4.10.2	3			
Humidity	3.11.4	4.10.4	4			
Salt fog	3.11.5	4.10.5		4		
Fungus	3.11.6	4.10.6	6			
Rain	3.11.7	4.10.7		3		
Dust	3.11.8	4.10.8	5			
Vibration/sinusoidal	3.11.9	4.10.9			1	
Shock	3.11.10	4.10.10			2	
Bench-handling	3.11.11	4.10.11			3	
6. EMI	3.12	4.11		2		
Safety checklist 2/	3.6	4.13.1		1		
7. Group D Reliability	3.13	4.12				1

1/ The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

2/ Safety checklist tests shall be performed only once during first article testing.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification and in the inspection requirement specified in the contract:

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.4
Nameplate and marking	3.5
Soldering Electrical and Electronic Assemblies	3.3

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4, and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each unit on contract shall be tested for conformance to the inspections in Table III. Lots shall be formed from units that pass this inspection. Each lot shall be subject to sampling inspection in accordance with Table IIIa. Lots in which samples exhibit any failures shall be screened for that failure prior to being subjected to Group B inspection.

4.5.1.1 Order of Test within Group A. Group A tests shall be performed in an order satisfactory to the government.

TABLE III. Group A Inspection

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical	3.1	4.13 4.14
System safety	3.6	4.13.1
Performance	3.10	4.9

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Table IIIa. Sampling For Group A Inspection

Lot Size	Sample Size
2 to 8	*
9 to 15	13
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	20

Notes:
1. '*' Indicates entire lot must be inspected
2. Acceptance number in all cases is zero.

4.5.2 Group B Inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to Group A inspection. These units shall be inspected for conformance to the inspection in Table IV. Each lot shall be subjected to sampling inspection in accordance with Table IVa. Lots in which samples exhibit any failures shall be screened for that failure prior to being subjected to Group C inspection.

TABLE IV. Group B Inspection

Inspection	Req. Para.	Insp. Para.
Interchangeability	3.7	4.8

Table IVa. Sampling For Group B Inspection

Lot Size	Sample Size
2 to 8	2
9 to 15	2
16 to 25	3
26 to 50	5
51 to 90	5
91 to 150	6
151 to 280	7

Note: Acceptance number in all cases is zero.

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4.5.3 Group C Inspection. Group C Inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met the Group A and B Tests.

4.5.3.1 Sampling for Group C Inspection. Three sample units shall be selected at random from every one-hundred units or fraction thereof produced. These samples shall be subjected to the inspections in table V. When first article is not required, the first samples shall be selected from the first quality conformance inspection production lot.

4.5.3.2 Order of Test within Group C. The order of test shall be in accordance with Table V, unless otherwise specified in the contract.

4.5.3.3 Group C Failures. Unless otherwise specified in the contract, failures which occur during Group C tesing must be reported to the cognizant procuring activity quality assurance representative within 48 hours of the failure. This may be accomplished telephonically or by any other means suitable of ensuring that the failure has been properly reported within the first 48 hours after the failure. The contractor shall immediately commence to identify the fundamental cause of failure and provide, within 30 days of failure, to the cognizant procuring activity quality assurance a detailed written failure report identifying the actual cause of failure and all effective corrective actions the contractor will take to preclude a reoccurrence of the failure. The government reserves the right to stop further acceptance of units which were manufactured using the same processes, procedures, and materials as the sample unit which failed. In addition, the government reserves the right to require the contractor to resubmit any failed unit(s), as well as any units they represent, to Group A and B testing prior to resuming acceptance of units on contract.

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TABLE V. Group C Inspection

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Altitude	3.11.1	4.10.1
High temperature	3.11.2	4.10.2
Low temperature	3.11.3	4.10.3
Humidity	3.11.4	4.10.4
Dust	3.11.8	4.10.8
Fungus	3.11.6	4.10.6
Subgroup 2		
EMI	3.12	4.11
Salt fog 1/	3.11.5	4.10.5
Rain	3.11.7	4.10.7
Subgroup 3		
Vibration, sinusoidal	3.11.9	4.10.9
Shock	3.11.10	4.10.10
Bench handling	3.11.11	4.10.11

1/ The equipment shall be thoroughly washed, dried and refurbished after their inspection before proceeding with subsequent inspection.

4.5.4 Group D Inspection. This inspection shall consist of the test specified in table VI and shall be performed on units from lots which have been subjected to and met group A and group B inspections.

4.5.4.1 Sampling for group D inspection. Two samples of AN/TPX-46A(V)7 shall be selected at random for group D inspection from every 50 units, or fraction thereof, produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

TABLE VI. Group D inspection

Inspection	Requirement paragraph	Test paragraph
Reliability	3.13	4.12

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4.5.4.2 Group D Failures. Unless otherwise specified in the contract, failures which occur during Group D tesing must be reported to the cognizant procuring activity quality assurance representative within 48 hours of the failure. This may be accomplished telephonically or by any other means suitable of ensuring that the failure has been properly reported within the first 48 hours after the failure. The contractor shall immediately commence to identify the fundamental cause of failure and provide, within 30 days of failure, to the cognizant procuring activity quality assurance representative a detailed written failure report identifying the actual cause of failure and all effective corrective actions the contractor will take to preclude a reoccurrence of the failure. The government reserves the right to stop further acceptance of units which were manufactured using the same processes, procedures, and materials as the sample unit which failed. In addition, the government reserves the right to require the contractor to resubmit the failed unit and all units it represents to Group A and B testing prior to resuming acceptance to units on contract.

4.5.4.3 Reinspection of group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

4.6 Burn-in/Pre-conditioning. Each of the major components of AN/TPX-46A(V)7 shall be subjected to vibration and electrical preconditioning. The preconditioning testing cycle shall consist of a four-hour period during which the chamber achieves a -45°C (-50°F) temperature (equipment not operating) followed by a four-hour period during which the temperature rises to +50°C (+124°F) (equipment operating). During the equipment operating period, the equipment shall be mechanically cycled periodically through its various phases of operation, as specified in the contract. Each equipment will be vibrated 10 minutes each operating hour at 1.25g $\pm$ 10% peak acceleration at any nonresonant frequency between 20 and 30 Hz. When a failure occurs, the equipment shall be repaired and preconditioning shall continue until at least three preconditioning cycles have been completed, of which the last cycle shall be failure free. A failure is any condition which results in a malfunction in the equipment. Preconditioning time and failures shall be recorded for each equipment. After the preconditioning and prior to electrical acceptance inspection, the equipment shall not be repaired, aligned, cleaned or adjusted except to the extent possible by use of the external operating controls (see 3.9).

4.7 Weight and dimensions. The major components of AN/TPX-46A(V)7 shall be weighed on a calibrated scale and measured to determine compliance with weight and dimension requirements of table I (see 3.8)

4.8 Interchangeability. The mechanical dimensions of electrical performance of parts shall be physically and electrically interchangeable among the units. Mechanical dimensions and electrical performance required by the drawings shall be measured to determine compliance with Requirement 7 of MIL-STD-454 (see 3.7).

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4.9 Performance test. The AN/TPX-46A(V)7 shall be subjected to performance and electrical inspection in accordance with 5071814 and SM-A-787975 (see 3.10).

4.10 Service conditions. The service condition tests shall be performed as detailed below. Each major component of AN/TPX-46A(V)7 subjected to these shall have passed group A tests and shall be resubmitted to and pass these group A tests after completion of all service condition tests. Where operation of the test item is required during any service condition test, the item shall be required to pass an operational test of 4.9.

4.10.1 Altitude. The AN/TPX-46A(V)7 shall be subjected to the test of MIL-STD-810, Procedure 1, Method 500.2. The AN/TPX-46A(V)7 shall be tested at 40,000 Ft (12,192m) non-operating for one hour. The operating altitude in Step 7 shall be 10,000 Ft (3,048m). Inability of AN/TPX-46A(V)7 to meet the requirements of 3.11.1 shall constitute failure of this test.

4.10.2 High temperature. The AN/TPX-46A(V)7 shall be subjected to the test of MIL-STD-810, Method 501.2, Procedure II. In Step 7 the temperature shall be 160°F (71°C). In Step 10 lower the temperature to 125°F (52°C), initiate operation of the equipment and at the end of a 30 minute period or when temperature stabilization occurs (whichever occurs first), the equipment shall show no degradation in performance (see 3.11.2).

4.10.3 Low temperature. The AN/TPX-46A(V)7 shall be subjected to the test of MIL-STD-810, Method 502.2, Procedure II. The temperature in Step 1 shall be maintained for a period of not less than 2 hours following stabilization of the test item. Following the 2 hour stabilization period, raise the chamber temperature to the low operating temperature (Step 4) of -50°F (-45.6°C) and maintain until temperature stabilization of the test item is reached. In Step 5, initiate operation of the equipment and at the end of 30 minute period or when temperature stabilization occurs (whichever occurs first), the equipment shall show no degradation in performance (see 3.11.3).

4.10.4 Humidity. The AN/TPX-46A(V)7 shall be subject to the test MIL-STD-810, Method 507.2, Procedure II. Prior to measurements, excess surface water may be removed from the equipment by wiping external surfaces only. In Step 6, operation of 4.9 measurements shall be performed during the last 5 hours of the last cycle (see 3.11.4).

4.10.5 Salt fog. The AN/TPX-46A(V)7 shall be tested in accordance with Method 509.2, Procedure 1, of MIL-STD-810. Inability of the AN/TPX-46A(V)7 to meet the requirements of 3.11.5 shall constitute failure of this test.

4.10.6 Fungus. The AN/TPX-46A(V)7 shall be subjected to the test of Method 508.3 of MIL-STD-810. There shall be abundant growth colonization on 50 percent or more of the control item after 14 and 28 days. No cleaning of the AN/TPX-46A(V)7 is permitted for 72 hours prior to the fungus test. Handling,

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prior to and during testing, shall be accomplished without contamination of the AN/TPX-46A(V)7. Inability of the AN/TPX-46A(V)7 to meet the requirements of 3.11.6 shall constitute failure of this test.

4.10.7 Rain. The AN/TPX-46A(V)7 shall be tested in accordance with Method 506.2, Procedure 1, of MIL-STD-810. The AN/TPX-46A(V)7 shall be operated during the last 10 minutes of each 30 minute test period. In each 30 minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period (see 3.11.7).

4.10.8 Dust (fine sand). The equipment shall be subjected to the test of Method 510.2, Procedure 1, of MIL-STD-810, except that the air velocity shall be 200 ±100 ft/min (1±0.5 m/sec) in Step 2. Step 3 shall proceed immediately after temperature and humidity are stabilized. The test item shall be reoriented every 1.5 hours during Step 4. If the time required to reorient the test item exceeds 5 minutes, it shall not be included in the test time of 6 hours (see 3.11.8).

4.10.9 Vibration, sinusoidal. The equipment shall be subjected to the test of Method 514.3, Category VIII, Figure 514.3-6, Paragraphs 4.5.1.1 and 4.5.1.3, Procedure 1, of MIL-STD-810. Test level shall be ±1.5g from 5.5 to 500 Hz (non-operating). Total vibration time shall be 3 hours, including resonance search time. Resonant dwell is not required (see 3.11.9).

4.10.10 Shock. The equipment shall be subjected to the test of Method 516.3, Procedure 1, of MIL-STD-810. The shock pulse shall be at a non-operating peak value of 20g's for a duration of 18 ms and an operating peak value of 9g's for a duration of 30 ms. (See 3.11.10)

4.10.11 Bench-handling. The equipment shall be subjected to the test of Method 516.3, Procedure VI, of MIL-STD-810. (See 3.11.11)

4.11 Electromagnetic interference. The AN/TPX-46A(V)7 system and the individual components shall be tested for compliance to the requirements of Notice 4 to MIL-STD-461A enumerated in 3.12. Testing shall be conducted in accordance with the methods and procedures of Notice 3 to MIL-STD-462 and the government approved Electromagnetic Interference Test Plan. Particular attention shall be paid to the requirements and intent of paragraph 4.2.7 (including it's subparagraphs on pages 26 through 28) of Notice 3 to MIL-STD-462 (Arrangement and Operation of Test Sample). (See 3.12)

4.12 Reliability. To determine compliance with 3.13, the equipment shall be tested in accordance with Plan IIIC of MIL-STD-781. The test level shall be in accordance with 4.12.1.

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4.12.1 Reliability test conditions. The reliability test desired above shall be conducted under conditions prescribed by Test Plan IIIC, MIL-STD-781, except that vibration shall be at 1.25g's. Voltage cycling shall be excluded and on-off cycling shall be 23 hours on and 1 hour off. The following requirements of MIL-STD-781 shall also apply:

a. Detail reliability test procedures (5.1.3).

b. Inspection (5.2).

c. Test facilities (5.4).

d. Selection of equipment (5.5.1).

e. Installation of equipment (5.5.2).

f. Failure categories (5.6.1).

g. Verifying repair (5.8).

4.13 Visual and mechanical. The AN/TPX-46A(V)7 shall be examined for the compliance with the equipment drawings, workmanship standards of MIL-STD -252, MIL-STD-2000, and marking requirements. In addition, AN/TPX-46A(V)7 shall be visually inspected to determine conformance to the requirements of 3.6.

4.13.1 Safety checklist. Using SEL 1183 checklist verify that the requirements of 3.6 are met.

4.13.2 Preshipment Inspection. The government reserves the right to perform any test or have the contractor perform any test or inspection the government deems necessary to ensure that the equipment and supplies conform to all contract and applicable specification requirements.

5. PACKAGING.

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity.

6. NOTES.

6.1 Intended use. Interrogator Set AN/TPX-46A(V)7 is intended for use with the Air Defense Radar Systems to provide the operator with a rapid and cryptographically secure means of identifying friendly aircraft.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of specification and any amendment thereto.

b. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

c. Marking and shipping samples.

d. Place of final inspection.

e. Number of first articles to be submitted for approval (see 3.1).

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, AN/TPX-46A(V)7. The contractor should apply for nomenclature in accordance with the applicable provision in the contract (see 1.1).

6.4 Subject term (key word) listing.

Connector, Program
Control
Foe
Friend
Identification
IFF
Signal Processor

Custodians:
Army - CR

Preparing Activity:
Army - CR
Project 5895-A426

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-49394A(CR)	2. DOCUMENT DATE (YYMMDD) 2 October 1991

3. DOCUMENT TITLE
INTERROGATOR SET AN/TPX-46A(V)7

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander, U.S. Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial 908-532-5851 (2) AUTOVON DSN-992-5851
c. ADDRESS (Include Zip Code) ATTN: AMSEL-ED-TM (Stdzn Ofc) Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340

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